

ARNAB DAS

Software Engineer — Voice AI, Generative AI & Full Stack

Kolkata / Hyderabad, India (open to remote) | +91-6296540769 | imarnab01@gmail.com | linkedin.com/in/arnab-das-2a5039182

SUMMARY

Software Engineer with 4 years building production AI systems and full-stack platforms. Lead engineer on a multi-channel, multilingual Voice AI platform delivering sub-400ms response latency and 97% transcript accuracy across English, Arabic and French for enterprise airline and contact-centre clients. Depth in real-time voice and telephony (LiveKit, WebRTC, SIP), LLM/RAG/agent systems and computer vision, backed by end-to-end ownership of a 17-repo multi-tenant SaaS platform on Azure. Delivered a 164x performance gain on a Government of India face-authentication system and a production data portal for a central bank. IIT Roorkee M.Tech. Serving notice — available to join within 15 days.

SKILLS

Languages: Python (3.10+), TypeScript, JavaScript (ES6+), C++, SQL

AI & ML: Azure OpenAI Realtime API, GPT-4o, LangChain, RAG, agentic systems, Pinecone, ChromaDB, FAISS, vector search, Whisper, YOLOv8, DeepFace (ArcFace), InsightFace, CLIP, BLIP, Piper TTS, FHIR R4

Real-Time & Telephony: LiveKit, WebRTC, WebSockets, SIP (Twilio Elastic SIP Trunk, Genesys AudioHook, CZentrix, MITEL), Azure Communication Services, STT/TTS pipelines

Backend: FastAPI, Node.js, Express.js, Django, REST, GraphQL, gRPC, Celery, Kafka, Redis Pub/Sub, OAuth 2.0

Frontend: React 18/19, Next.js 14/15, Redux Toolkit, React Query, React Native, Angular.js, Tailwind CSS, responsive design

Data & Infrastructure: PostgreSQL, MongoDB, Redis, MySQL, SQLite, Supabase, Docker, Nginx, systemd, CI/CD, Azure (Container Apps, VMs), GCP (P100/P4 GPU), Alembic, Playwright, FFmpeg

Security: JWT (RS512/HS256), HMAC webhook signature verification, RBAC design, API-key and secrets management, TLS/certificate troubleshooting

EXPERIENCE

Software Engineer — AlonOS

Hyderabad, India | Feb 2025 - Present

Saudia Voice-AI Platform — Enterprise Voice AI System (Microsoft-Sponsored)

- Delivered an end-to-end enterprise voice AI platform for Saudia Airlines at sub-400ms response latency and 97% transcript accuracy across English, Arabic and French, integrating Azure OpenAI Realtime API with Twilio telephony and LiveKit real-time transport.
- Cut first-audio delivery latency roughly 6x (3,031ms to ~500ms) by diagnosing and fixing a LiveKit agent initialisation race condition, measurably improving perceived call quality.
- Diagnosed and resolved a rogue-worker production incident causing silent task leaks in the distributed session manager, traced across services using distributed log correlation and process-level forensics.
- Enabled complete end-to-end booking workflows through natural voice conversation by exposing 10+ Saudia APIs (flight search, booking, modification) as voice-callable tools via OpenAI function calling.
- Reduced redundant data collection 78% by architecting persistent conversation continuity with Redis-backed context preservation and priority context injection across disconnected calls.
- Reduced vector search latency 200-300ms by migrating conversation context retrieval from ChromaDB to Pinecone, lowering P95 session-lookup time under concurrent load.
- Reduced customer service tickets 52% with multi-modal booking confirmations (automated email and SMS); achieved 97% uptime through WebSocket auto-reconnect, heartbeat monitoring and full session lifecycle management.

UniWeave Enterprise Platform — Full-Stack Platform Engineering, Security & Telephony

- Onboarded independently to a 17-repo enterprise multi-tenant SaaS codebase with zero handoff documentation, becoming primary technical owner within two weeks; authored a 920-line Platform Engineering Standards document formalising coding conventions, test requirements and deployment procedures platform-wide.
- Resolved a critical cross-org data leak as anchor engineer on an RBAC hardening sprint that closed 13 security vulnerabilities, ensuring complete tenant data isolation for enterprise clients.
- Led a platform-wide authentication redesign and backend consolidation: merged 4 independently-deployed services into a single consolidated API (245+ files, verified byte-for-byte), migrated JWT signing from RS512 to HS256 with zero-downtime dual-key acceptance (algorithm-confusion-resistant, keyset-pinned verification), and ported a Python SIP tokenizer and dispatcher to TypeScript.
- Closed the epic's highest-severity finding — a webhook ingress with no signature verification capable of forcing room dispatch or deletion on the shared chokepoint for every call channel — plus 2 undocumented unauthenticated toll-fraud-capable routes and a hardcoded API key live in source.
- Ran a full black-box regression pass (12-folder Postman suite written from live controller source) before production sign-off, catching 6 real product bugs invisible to unit tests, including an auto-fill defect that would have broken agent creation for every non-super-admin user.
- Root-caused a persistent SIP "connects but no audio" defect to a missing RTP error check in the open-source livekit/sip library, confirmed by source review across every release from v1.0 to v1.7; validated the platform's own SIP architecture within 48 hours by designing, building and deploying a Twilio ↔ self-hosted LiveKit proof-of-concept (5-service Docker Compose stack, real Elastic SIP Trunk, successful outbound call).
- Delivered a full-lifecycle Salesforce CRM integration for the real-time voice platform — OAuth 2.0 client-credentials auth with Redis-cached tokens, idempotent Case/Contact sync, identity-verified write-gating preventing record misattribution — shipped across 4 production releases with a regression pass catching 4 live-call race conditions; later genericised to a single feature flag as the architecture evolved.
- Rebuilt live-call monitoring twice in one sprint: a 5-phase security remediation closing an access-control gap that let any authenticated user view or force-disconnect every live call across every organisation (2 unauthenticated endpoints removed, single-use HS256-signed access tickets introduced, per-connection organisation filtering enforced), then an authoritative real-time monitoring view on the video infrastructure's native room-listing API.
- Designed and implemented CZentrix SIP integration enabling verified outbound telephony, validated end-to-end with a 36-second production phone call and integrated with the platform's real-time communication layer.
- Executed a 6-phase Pulse Restoration Programme delivering +5,312 lines of production code in 3 days to restore full platform functionality ahead of a \$5M enterprise client demo and executive presentations.
- Built 2 full-platform Playwright E2E suites documenting 58 reproducible bugs across the 17-repo codebase, establishing the first systematic test coverage baseline and enabling structured defect triage.

- Led a 255-file design token migration standardising the platform-wide design system, and a 50+ page React Query migration replacing ad-hoc data fetching with structured server-state management.
- Architected envsubst-based secrets management for Azure infrastructure — environment variables as single source of truth across 5 Container Apps and 2 VMs — enabling zero-downtime key rotation across the deployment fleet.
- Enforced code quality through systematic PR review for a 5-person engineering team, catching 3 critical defects before production merge; defined technical evaluation criteria and screened 28+ candidates across 3 engineering hires.

IntelliVision Platform — Enterprise Computer Vision Platform (Full-Stack, In-House Product)

- Architected and delivered a 9-analysis-type computer vision platform (people counting, crowd detection, parking analysis, wildlife detection, number plate recognition, PPE compliance, pothole detection, food waste estimation, room readiness) with a FastAPI backend, Celery async task processing and a React/TypeScript frontend with Redux state management.
- Implemented BYOK (Bring Your Own Key) multi-provider LLM architecture supporting 5 providers (OpenAI, Anthropic, Google, Grok, Azure AI Foundry) with Fernet-encrypted key storage, dynamic client factory pattern and connection testing API, eliminating central API cost bottlenecks.
- Built a webcam-based live demo achieving sub-2-second analysis latency at 1 FPS with client-side RecordRTC recording (30 FPS WebM), enabling real-time CV demonstrations without pre-recorded uploads.
- Delivered complete UI/UX modernisation: 46 production-grade mockups, a glassmorphism design system across 20+ components, 12 domain-specific result visualisations and role-based navigation architecture.
- Deployed production infrastructure on Azure VM with 3 isolated ML microservice environments, Let's Encrypt SSL, Nginx reverse proxy and systemd management.
- Migrated the database layer from SQLite to PostgreSQL with Alembic migration infrastructure and connection pooling; designed a microservices architecture proposal for long-term scalability.
- Authored a 23,500-word platform knowledge transfer document covering architecture decisions, API contracts, deployment procedures and operational runbooks, enabling full team onboarding across the stack.

IntelliVideo Platform — Video Intelligence MVP (In-House Product)

- Architected a complete video intelligence platform with 31 REST APIs covering multimodal search, dynamic video compilation and advanced person identification — full MVP delivered in a single development sprint.
- Built an advanced person identification system achieving 95%+ accuracy using a dual face recognition pipeline (DeepFace ArcFace + InsightFace embeddings) with multi-strategy preprocessing, async Celery task architecture and database-persisted match results, supporting missing person investigation workflows.
- Achieved 40% memory optimisation integrating 5 AI models (Whisper, BLIP, CLIP, YOLOv8, SentenceTransformer) with lazy loading architecture, FAISS vector store and GPU-accelerated inference on GCP Tesla P4.
- Deployed on GCP with GPU acceleration, systemd management, Nginx reverse proxy and enterprise security (token-based auth, 6-layer file validation, intelligent rate limiting), achieving 99.5% uptime.

IntelliConverse & Genesys Audio Connector — Voice AI Platform Integration

- Delivered a production-ready Genesys Audio Connector enabling seamless integration between Genesys Cloud telephony infrastructure and UltraVox voice AI for enterprise call centre workflows.
- Implemented bidirectional audio streaming with sample rate conversion (8kHz PCMU to 48kHz PCM), X-API-KEY validation and RFC 8941 structured fields signature verification.
- Authored comprehensive 9-microservice architecture documentation spanning 8 codebases across 4 technology stacks (Java/Spring Boot, Node.js/Express, Python/FastAPI, Next.js 15).

IntelliAvatar — Real-Time Conversational AI Avatar (R&D)

- Evaluated 4 SOTA frameworks (MuseTalk, LivePortrait, AIAvatarKit, TEN Framework) for real-time photorealistic avatar generation targeting airport kiosk deployment; validated LivePortrait as the recommended visual engine and documented the critical GPU requirement (CPU >300s/frame vs GPU 40-100ms/frame, ~3,000x delta).
- Implemented the complete backend AI pipeline: Groq LLM (Llama-3.3-70B), Piper TTS (22,050Hz/16-bit), espeak-ng phonemizer with 50+ IPA-to-viseme mappings, achieving 3.5-5s end-to-end conversational latency with synchronised lip-sync.
- Implemented a SELECT FOR UPDATE SKIP LOCKED queue pattern for concurrent task dequeuing, preventing phantom reads and task starvation under sustained avatar request load.
- Modelled GPU compute cost for VL-JEPA visual-language architecture training, projecting a \$12K-\$60K infrastructure requirement to guide platform R&D investment decisions.
- Redesigned architecture from LiveKit/Pipewait to streamlined WebSocket communication, eliminating 3 heavy framework dependencies and enabling self-hosted LLM integration on AlonOS H200 infrastructure.

Cross-Project Technical Contributions

- **IndiaAI Face Authentication Challenge (Government of India):** achieved a 164x performance improvement (90s to under 1s per submission) by optimising face matching from O(n) sequential comparison to vectorised NumPy cosine similarity across 5,000+ stored records using FaceNet; migrated the backend from Flask to FastAPI with dual-layer security (API key + JWT) and built a React/Redux Toolkit admin dashboard with real-time duplicate detection statistics.
- Architected a distributed task orchestration engine using Kafka, Redis Pub/Sub and PostgreSQL, processing 500+ concurrent voice sessions at 99% task completion rate and 3x throughput improvement.
- Delivered the Central Bank of Oman Data Collection Portal frontend (~8,000 lines TypeScript/React) in a 28-hour sprint — multi-role authentication across 4 user roles, a SheetJS Excel validation engine with 15+ regulatory compliance rules, deployed to a production VM.
- Diagnosed a Redis SSL/TLS kill-loop bug (60-second connection termination cycle) in production, tracing root cause to a TLS handshake configuration mismatch and restoring full cache layer reliability.
- Built a configurable Generic REST Caller microservice reducing system coupling 60% through single-endpoint design with dynamic request/response mapping supporting 15+ external APIs.
- Achieved 93% variance reduction (\$42,817 to ~\$3,000) in the Pourri POC Amazon deduction reconciliation by validating a 2,118-line API codebase's aggregation logic, UPC mapping and classification pipeline.
- Authored CI/CD pipeline specifications across 10 repositories, standardising build, test and deployment workflows and reducing integration errors.
- **UniHealth NHCX (National Health Claims Exchange):** built a complete FHIR R4 patient data ingestion and normalisation pipeline achieving zero errors across all 4 official test PDFs, parsing and transforming unstructured healthcare documents into FHIR-compliant JSON resources.

Software Engineer — ARThink AI

Hyderabad, India | Jan 2024 – Jan 2025

- **QuadzAI — Intelligent Chatbot Platform:** built Python vector search APIs using ChromaDB and Pinecone for semantic search, improving result accuracy 25%, and integrated OpenAI LLMs for context-aware responses; developed Node.js APIs for web scraping, PDF text extraction and MongoDB integration, boosting indexing speed 40%.
- **OnFit AI — Healthcare PWA & Mobile App:** designed Socket.IO real-time communication infrastructure at 99% uptime across mobile platforms; built a two-user WebRTC video conferencing PoC for virtual consultations; implemented Firebase Cloud Messaging and web push, upgrading FCM to HTTPS v1 and reducing data breach risk 30%.
- Delivered enterprise app features including advanced file sharing with PDF previews, enhanced chat (reply/edit/delete), real-time patient dashboards, metadata-driven UI and Google Fit API integration; managed Play Console releases via Expo EAS, cutting manual deployment effort 50%.
- **Telangana Government Widget Programme:** designed scalable backend architecture for real-time tracking of government social schemes across rural regions, reducing administrative delays 40%; built high-performance data APIs supporting multi-platform visibility and real-time policy reporting.

Junior Software Engineer — SRM Software Inc

Remote | Jan 2023 – Sep 2023

- Automated HR workflows with Groovy scripting, reducing manual tasks 60%; maintained 99.9% uptime on production-grade enterprise tools and enhanced an Angular-based frontend, improving client satisfaction scores 25%.
- Earlier internships (2020-2022):** Tradebuilder Technologies — Next.js/Node.js e-commerce dashboards and real-time order management APIs (30% downtime reduction). Outshade Digital Media — secure FinTech payment management APIs (35% fewer processing errors). Cartel101 — hospital management frontend and HIPAA-compliant medical record APIs (25% faster page loads). Adwaita Educare — OTP registration with JWT auth and Razorpay integration (30% higher onboarding success). Expidense — responsive expense management system and real-time reporting APIs.

EDUCATION

Integrated M.Tech, Geophysical Technology — Indian Institute of Technology (IIT) Roorkee, 2023
AISSE (PCM + Computer Science), Jawahar Navodaya Vidyalaya, Bengaluru Urban, 2018 — 91.4%

LEADERSHIP & EARLY VENTURES

- Cell Secretary, Web Development Cell, NSS IIT Roorkee** (2020-21) — technical head of a 14-member engineering team delivering 5+ mission-critical web platforms for institutional and social service initiatives; organised annual workshops for 150+ students.
- Technical Head, National Social Summit, IIT Roorkee** (2020-21) — led the technical team delivering an event platform for 3,000+ participants with registration, live streaming and real-time communication infrastructure.
- Founding Engineer (Front-End), Zixwer** (2020-22) — led front-end architecture and product development across educational, construction and travel management platforms, from zero to production. **Lead Front-End Developer, Bechho** (2020) — sole front-end developer delivering a hyperlocal business management platform from scratch.

LANGUAGES

Bengali (native) | English (professional working proficiency) | Hindi (professional working proficiency)